

📖 FULL OS PATCHING GUIDE

🔒 WHAT IS OS PATCHING?

OS Patching is the process of applying updates or fixes to an operating system. These updates may include:

- Security patches (to fix known CVEs)
- Bug fixes
- Performance enhancements
- Feature improvements

They are released by OS vendors like Red Hat, Oracle, Canonical (Ubuntu), etc.

🔧 WHY DO WE PERFORM PATCHING?

1. 🛡️ Security:

- Fix vulnerabilities that could be exploited (e.g., CVE-2024-29813)
- Protect the system from malware, ransomware, etc.

2. ✅ Compliance:

- Required by IT audits (PCI-DSS, HIPAA, ISO 27001)
- Prevent non-compliance penalties

3. ⚙️ Stability:

- Fixes kernel bugs and performance issues
- Reduces system crashes and downtime

4. 📈 Performance:

- Kernel and package-level optimizations
- Faster memory, I/O handling, etc.

5. ☐ Compatibility:

- Prepares the system for new software or drivers
- Avoids dependency issues in future upgrades

🛠️ PRE-PATCHING CHECKLIST

1. Backup Critical Data:

```
mkdir -p /backup/app-server/etc  
rsync -av /etc /backup/app-server/etc/  
mysqldump -u root -p appdb > /backup/app-server/appdb.sql
```

2. Note Current System State:

```
hostnamectl  
uname -r  
uptime  
df -h  
free -m
```

3. Save Package Versions:

```
rpm -qa > /root/package_list_before_patch.txt
```

4. Check Repository Access:

```
yum repolist  
ping internal.repo.local
```

5. Notify Teams:

- Application owner
- DBA
- Network/infra
- Raise Change Request (CR)

PATCHING PROCEDURE (WITH OUTPUT)

Step 1: Clean YUM Cache

```
yum clean all  
Output:  
0 files removed
```

Step 2: Check for Updates

```
yum check-update  
Output:  
kernel.x86_64 4.18.0-408.el8 rhel-8-baseos-rpms  
openssl.x86_64 1:1.1.1k-7.el8 rhel-8-baseos-rpms
```

Step 3: Apply All Patches

```
yum update -y  
Output (example):  
Upgraded:  
kernel-4.18.0-408.el8  
openssl-1.1.1k-7.el8  
Complete!
```

Step 4: Reboot the Server

```
reboot
```

Step 5: Verify New Kernel After Reboot

```
uname -r  
Output:
```

4.18.0-408.el8.x86_64

Step 6: Save Post-Patch Package List

```
rpm -qa > /root/package_list_after_patch.txt
```

Step 7: Append Patch History

```
echo "Patched on: $(date)" >> /root/patch-history.log
```

🔍 POST-PATCHING CHECKLIST

1. Re-check System Health:

```
df -h
```

```
free -m
```

```
uptime
```

2. Validate Services:

```
systemctl status nginx
```

```
systemctl status mysqld
```

3. Test Application/UI/API:

- Visit app URL: <https://app-prd-01/>

- Perform a sample transaction

- Review logs: `tail -f /opt/app/logs/app.log`

4. Re-run Vulnerability Scan:

Use Nessus or Qualys

- Confirm CVEs are fixed

- Attach scan report to ticket

5. Document Everything:

- Pre/post package list

- Kernel versions

- Screenshot of scan report

- Patch history

📁 FINAL AUDIT-READY SUMMARY

Task	Status
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Backup Taken	✓
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Prechecks Done	✓
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Patches Applied	✓
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System Rebooted	✓
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Postchecks Performed	✓
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Scan Revalidated	✓
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| Ticket/CR Closed | ✓ |

✓ CONCLUSION

OS Patching is critical in maintaining the health and security of any IT environment. A structured approach with proper planning, validation, and reporting ensures:

- Security of systems
- Compliance with standards
- High availability and performance
- Peace of mind during audits

Always remember the golden rule:

🔒 Backup → Patch → Validate → Document